

THYMAZOLE

DESCRIPTION: 7mm round tablet, white in colour with "Star" marking and "DUO 861".

COMPOSITION: Each tablet contains Carbimazole 5 mg.

PHARMACODYNAMICS: The normal function of the thyroid gland is to produce thyroid hormones, the chief being L-thyroxine (T₄), which is an amino-acid containing 4 atoms of iodine in each molecule. Its formation and release into the circulation are under the control of the pituitary thyrotrophic hormone (T.S.H.). Iodides in the normal diet are absorbed from the gastrointestinal tract, enter the blood stream and are trapped by the thyroid gland. No other tissue in the body has this property. The thyroid is capable of trapping 100 times the amount of iodide that would be expected to enter the gland by normal diffusion. In thyrotoxicosis the gland may collect up to 500 times the normal amount of iodide. Once the iodide is in the gland it is oxidised into iodine by an iodide peroxidase. As has been mentioned earlier, the level of production and release of thyroid hormones is regulated by the thyroid-stimulating hormone of the pituitary. In addition, to some extent thyroid activity is influenced by the level of blood iodide, the temperature of the blood and, to a very much lesser extent, by the nervous system. The thyroid can conserve iodine, and has the ability to liberate iodine for new synthesis from spare iodine-containing molecules. Unlike other hormones, thyroxine acts upon every tissue in the body and is not merely specific for certain organs. The concentration of it that reaches any cell in the body controls the speed with which that cell metabolises. Thyroxine is relatively slow-acting and may take several days to exert its effect but T₃ takes only a few hours to alter cell metabolism. Carbimazole is believed to exert its antithyroid effect inhibiting the iodination of thyrosine. It is also thought to have some action on the enzyme peroxidase which is required as a catalyst in the synthesis of thyroxine by the thyroid gland. It does not affect the uptake of iodine by the thyroid gland and this is of vital importance in the treatment of thyrotoxicosis with radio-active iodine or with a combination of radio iodine and Carbimazole and also in preparation of patients for operation

PHARMACOKINETICS:

Absorption and distribution: Carbimazole is absorbed from the GI tract as methimazole. It crosses the placenta and is present in milk during lactation. Carbimazole is hydrolysed in the blood to methimazole.

Excretion: Carbimazole is excreted in the urine, excretion being almost complete in 24 hours.

Plasma levels: Analysis for carbimazole in the blood serum is difficult because of the speed it is metabolised in plasma to methimazole. As such most studies are geared to the determination of its active metabolite, methimazole. It has been reported that after a single oral dose (60 mg), the patients' blood level appeared to fall into two groups, either those with a minimum concentration of methimazole between 30 and 60 minutes (in the range of 0.2 to 0.9 mcg/mL) or those whose maximum was in the range of 0.4-1.1 mcg/mL achieved in 2-3 hours.

INDICATIONS:

Thyrotoxicosis:

- Primary thyrotoxicosis
- Secondary thyrotoxicosis - toxic nodular goitre.

However, Carbimazole really has three principal applications in the therapy of hyperthyroidism:

- Definitive therapy - induction of a permanent remission
- Preparation of thyroidectomy
- Before and after radio-active iodine treatment

RECOMMENDED DOSAGE:

The great majority of patients suitable for Carbimazole can be treated as out-patients and can in fact carry on normal occupations and household duties.

Initial dosage of Carbimazole: It is customary to begin Carbimazole therapy with a dosage that will fairly quickly control the thyrotoxicosis and render the patient euthyroid, and later to reduce this. The usual initial dosage for adults is up to 60 mg per day given in divided doses. Thus:

Mild cases	20 mg	Daily in 8 hourly divided dosage
Moderate cases	40 mg	
Severe cases	40-60 mg	

Three factors determine the time that elapses before a response is apparent:

- The quantity of hormone stored in the gland. (Exhaustion of these stores usually takes about a fortnight).
- The gland secretory rate.
- The degree of inhibition of hormone synthesis achieved by Carbimazole.

If large stores of hormones are present, as in nodular goitre, response to Carbimazole may be delayed for several weeks or months, whereas in severe thyrotoxicosis, when very little hormone is stored, improvement may be detected within three to four days.

Maintenance dosage of Carbimazole: When symptoms are controlled the dosage should be reduced to a maintenance level, which will usually be between 10 and 15 mg daily. Experience has shown there is a wide variation of sensitivity to the drug and from time to time in a particular patient. For this reason patients should be seen monthly for the first year, and thereafter at three monthly intervals.

Duration of Carbimazole therapy: First, the time required to render a patient euthyroid depends very much on the type of case being treated. Toxic nodular goitres usually take very much longer (Also see **Precautions**). Second, once a remission has been secured maintenance dosage should be continued for at least twelve months, and up to two years of Carbimazole treatment may be required. Of course, if thyroidectomy is intended, it can be carried out once the euthyroid state is achieved with Carbimazole, which is then discontinued.

Change-over from thiouracils: When treatment with one of the thiouracils is replaced by Carbimazole therapy, one 50 mg tablet of methylthiouracil or propylthiouracil can be taken as equivalent to one 5mg tablet of Carbimazole.

ROUTE OF ADMINISTRATION:

Oral

CONTRAINDICATIONS: Compatibility: There are no reports that Carbimazole is incompatible with any other drugs or foods. Carbimazole should be given with the utmost caution, or not at all, if there is any degree of tracheal obstruction, as high dosage may produce thyroid enlargement and aggravate obstructive symptoms. Carbimazole may be given during pregnancy to a thyrotoxic patient but the smallest effective dose should be used and treatment should be discontinued three or four weeks before delivery. Infants should not be breast fed by mothers taking carbimazole.

Patient with a history of acute pancreatitis after administration of carbimazole or active metabolite, methimazole (thiamazole).

WARNING & PRECAUTIONS:

Delayed response to Carbimazole therapy: If no relief is obtained within three months, the possible causes are:

- Patients have failed to take their Carbimazole. This is the most common cause.
- Previous iodine therapy which has resulted in an increased hormone store within the gland.
- Inadequate dosage of carbimazole.

Preparation of thyrotoxic patients for surgery: Carbimazole is prescribed prior to thyroidectomy, and should then be given in sufficient dosage and for long enough to render the patient euthyroid.

Carbimazole may cause white cell disorders such as neutropenia and agranulocytosis, which may be fatal if treatment with carbimazole is not stopped promptly. These reactions usually occur during the first 3 months of therapy, and in most cases, are reversible on stopping treatment. Since agranulocytosis can develop very rapidly, periodic leucocyte counts

alone may not be effective in the early detection of these reactions.

There have been post-marketing reports of acute pancreatitis in patients receiving carbimazole or its active metabolite, methimazole (thiamazole). In case of acute pancreatitis, carbimazole or methimazole (thiamazole) should be discontinued immediately. Carbimazole or methimazole (thiamazole) must not be given to patients with a history of acute pancreatitis after administration of carbimazole or its active metabolite methimazole (thiamazole). Re-exposure may result in recurrence of acute pancreatitis, with decreased time to onset.

Women of childbearing potential and pregnancy

Women of childbearing potential have to use effective contraceptive measures during treatment.

The use of carbimazole or methimazole (thiamazole) in pregnant women must be based on the individual benefit/risk assessment. If carbimazole or methimazole (thiamazole) is used during pregnancy, the lowest effective dose without additional administration of thyroid hormones should be administered. Close maternal, foetal and neonatal monitoring is warranted.

INTERACTIONS WITH OTHER MEDICAMENTS: None Available

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Carbimazole and related drugs cross the placenta and are concentrated in the breast milk. During the later stages of pregnancy, it may be desirable to halve the usual dosage in order to reduce the risk of foetal goitre and hypothyroidism.

Carbimazole or methimazole (thiamazole) crosses the placenta but, provided the mother's dose is within the standard range and her thyroid status is monitored; there is no evidence of neonatal thyroid abnormalities. Studies have shown that the incidence of congenital malformations is greater in the children of mothers whose hyperthyroidism has remained untreated than in those who have been treated with carbimazole or methimazole (thiamazole).

However, cases of congenital malformations have been observed following the use of carbimazole or its active metabolite, methimazole (thiamazole) during pregnancy.

A causal relationship of these malformations, especially choanal atresia and aplasia cutis congenita (congenital scalp defects), to transplacental exposure to carbimazole and methimazole (thiamazole) cannot be excluded.

Therefore the use of carbimazole or methimazole (thiamazole) in non-pregnant women of childbearing potential should be based on individual risk/benefit assessment (see Warning and Precautions).

Cases of renal, skull, cardiovascular congenital defects, exomphalos, gastrointestinal malformation, umbilical malformation and duodenal atresia have also been reported. Therefore, carbimazole or methimazole (thiamazole) should be used in pregnancy only when propylthiouracil is not suitable.

If carbimazole or methimazole (thiamazole) is used in pregnancy, the dose must be regulated by the patient's clinical condition. The lowest dose possible should be used, and this can often be discontinued three or four weeks before term, in order to reduce the risk of neonatal complications.

The blocking-replacement regimen should not be used during pregnancy since very little thyroxine crosses the placenta in the last trimester.

Hyperthyroidism in pregnant women should be adequately treated to prevent serious maternal and foetal complications.

Carbimazole or methimazole (thiamazole) is able to cross the human placenta.

Based on human experience from epidemiological studies and spontaneous reporting, carbimazole or methimazole (thiamazole) is suspected to cause congenital malformations when administered during pregnancy, particularly in the first trimester of pregnancy and at high doses.

Reported malformations include aplasia cutis congenita, craniofacial malformations (choanal atresia; facial dysmorphism), exomphalos, oesophageal atresia, omphalo-mesenteric duct anomaly, and ventricular septal defect.

Carbimazole or methimazole (thiamazole) must only be administered during pregnancy after a strict individual benefit/risk assessment and only at the lowest effective dose without additional administration of thyroid hormones. If carbimazole or methimazole (thiamazole) is used during pregnancy, close maternal, foetal and neonatal monitoring is recommended (see section Warnings and Precautions).

Women of childbearing potential and pregnancy

Women of childbearing potential have to use effective contraceptive measures during treatment (see Section Warnings and Precautions).

Use in Lactation: Infants should not be breast fed by mothers taking Carbimazole.

SIDE EFFECTS:

Adverse reactions usually appear in the first few weeks of therapy and may consist of skin rash, urticaria, joint pains, fever, sore throat, anorexia, malaise, agranulocytosis and toxic neuropathy. The principal manifestations of poisoning are skin rash and leucopenia. Acute poisoning has not been reported.

SOC Gastrointestinal disorders

Frequency "Not Known": Acute pancreatitis

SYMPTOMS AND TREATMENT OF OVERDOSE:

Agranulocytosis is the most serious adverse reaction, and patients should be instructed to report the development of mouth ulcers or sore throat, fever, or rashes, since these may sometimes precede abnormal findings in the blood by several days. Thrombocytopenia, leucopenia, and aplastic anaemia have also been reported. Cross-sensitivity between antithyroid agents may occur.

Treatment of adverse effects: If blood disorders, such as agranulocytosis or bone-marrow depression occur, antithyroid agents should be immediately withdrawn and, if necessary, antibiotics and blood transfusions may be given.

PACK SIZE: Blister pack of 10 x 10's and 100 x 10's tablets in a box.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

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